November 16

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[The insulation degradation detection function enables the ground insulation resistance values of motors and cables to be checked using the voltage charged to a servo amplifier's DC link, allowing the signs of insulation degradation to be detected at high accuracy.
The ROBODRILL a-DiB se1ies provides a dedicated screen that allows you to operate the insulation degradation detection function using the screen]

[The ROBODRILL a-DiB se1ies provides a dedicated screen that allows you to operate the insulation degradation detection function using the screen]

Note:
"The INSULATION DEGRADATION DETECTION FUNCTION screen allows you to perform the following operations.

- **Display the results of executing the insulation degradation detection function**
- **Display guidance on the insulation degradation detection steps**
- **Configure various insulation degradation detection function settings… "**

General Cautions

a. Safety first.

b. Special care is need for handling any parts.

c. Mark on the any parts for assembling in the original position and direction when disassembling.

Required tools

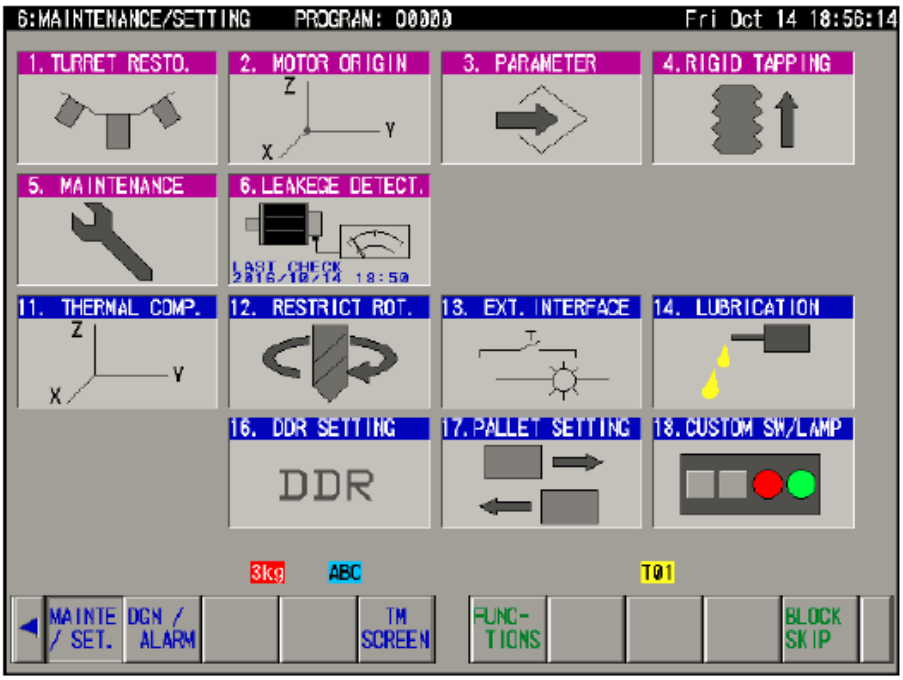
・General working tools.

1. Multimeter
2. Screw Drivers
3. Allen Key Sets
4. Spanners
5.

Working Procedure

Methods for Switching to Insulation Degradation Detection Screen

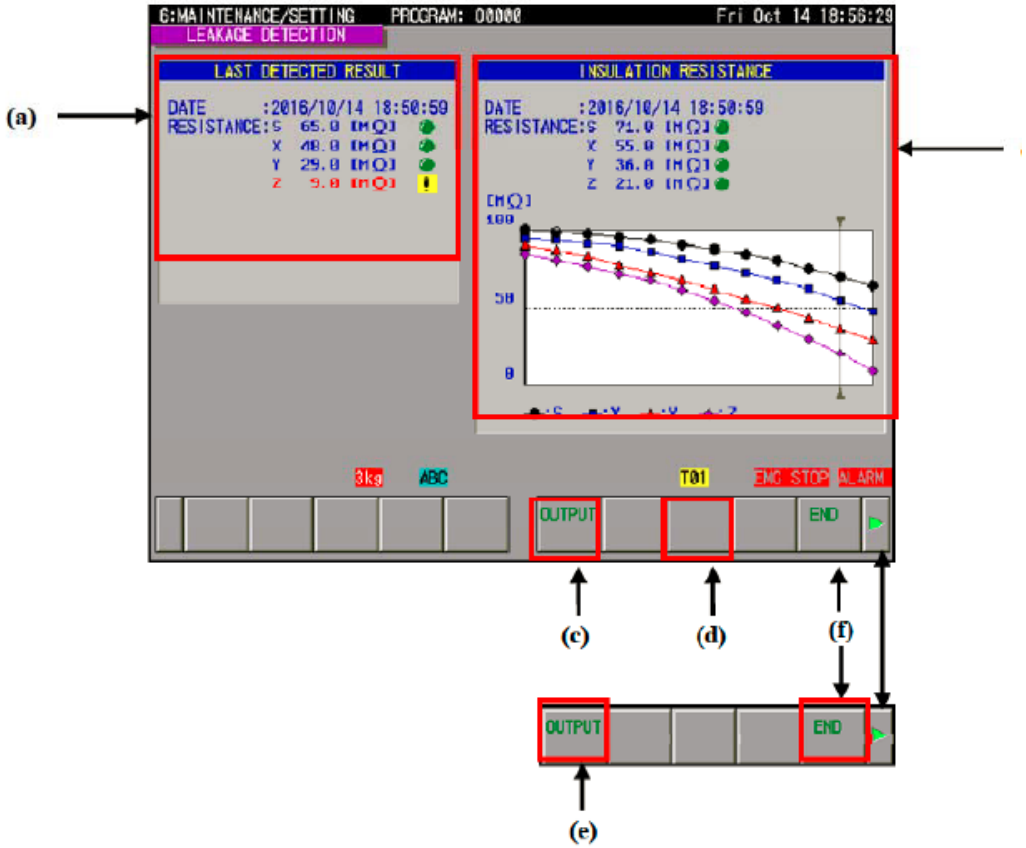
1) Press the [MAINTENANCE/SETTING] soft key on the QUICK screen, then display the MAINTENANCE/SETTING" screen



2) Press the <6> + <INPUT> keys. The dedicated screen for the insulation degradation detection function appears.

Principles of Insulation Degradation Detection Function Screen

INSULATION DEGRADATION DETECTION FUNCTION SCREEN consists of the following.



(a) Final detection results

Displays the date and time of the last execution of the insulation degradation detection function and the then measured insulation resistance value.
For details, see "47.3.2 checking the Execution Results of the Insulation Degradation Detection Function".

(b) Changes of insulation resistance value
Saves the execution results of the insulation degradation detection function for up to 12 times and shows them in a graph. By hovering the cursor over the graph, you can check the details of each data item plotted on the graph.
For details, see "47.3.2 checking the Execution Results of the Insulation Degradation Detection Function".

(c) [DETECT] soft key
Displays the "EXECUTE INSULATION DEGRADATIONDETECTION" window.
You can execute the insulation degradation detection function by operating the EMERGENCY STOP button in accordance with the guidance displayed.
For details, see "47.3.1 executing the Insulation Degradation Detection Function."

(d) [CHANGE SETTINGS] soft key
Displays the "SETTINGS" window. You can configure the following settings related to the insulation degradation detect

Execution of insulation degradation detection

The conditions to execute the insulation degradation detection function and save execution results are set.
For details, see "47.3.3 Settings Related to the Execution of the Insulation Degradation Detection Function."

Notification of execution of detection (notification method, notification interval)

By outputting an alarm or operator message when a ce1tain time has passed after the last execution of the insulation degradation detection function, the need to execute insulation degradation detection can be notified so that the function executed at regular intervals.

For details, see "47.4 NOTIFYING EXECUTION OF INSULATION DEGRADATION DETECTION FUNCTION. II

Execute when switching the power on

This function cannot be used on the ROBODRILL a-D iB se1ie s.

Warning level

The warning level (the insulation resistance value for dete1mining that insulation degradation has occurred can be set.
For details, see "47.6 WARNING LEVELS."

When the setting is complete, press the [END] soft key to close the window.
At that time, if any item has been changed the following soft keys appear Select and press one of the soft keys.
[YES] : Saves the change and close the "SETTINGS" window. [NO]: Closes the "SETTINGS" window without saving the change.
[CANCEL]: Continues the change of settings without closing the "SETTINGS" window.

(e) [OUTPUT] soft key

This soft key enables the execution results of the insulation degradation detection function to be output to a memory card or USB memory.
For details, see "47.7 OUTPUT OF EXECUTION RESULTS

(f) [OUTPUT] soft key

This soft key closes the INSULATION DEGRADATION DETECTION FUNCTION screen and returns the screen display to the "6: MAINTENANCE/SEITING" screen.

47.3 EXECUTION OF INSULATION DEGRADATION DETECTION FUNCTION

This section describes the problem to execute the insulation degradation detection function using the INSULATION DEGRADATION DETECTION FUNCTION screen and check the execution results

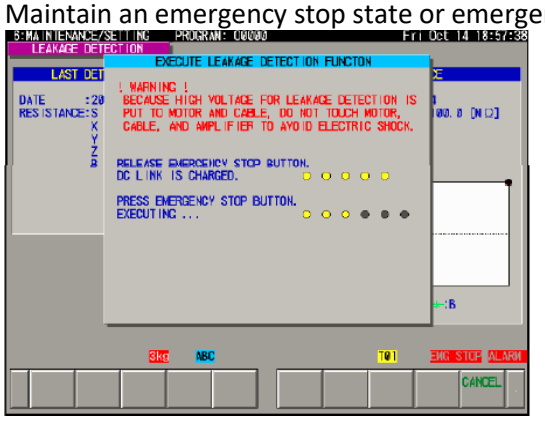
47.3.1 Execution of Insulation Degradation Detection Function

On the "EXECUTE INSULATION DEGRADATION DETECTION" window, you can execute the insulation degradation detection function

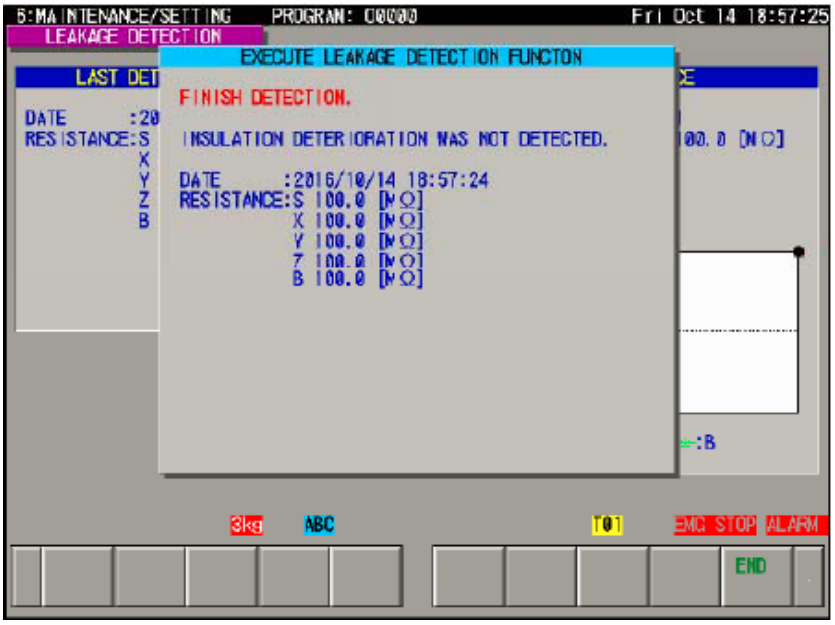
(1) Press the [DETECT soft key the following window appears]

(2) Press the [EXECUTE] soft key.

(3) Operate the EMERGENCY STOP button according to the guidance displayed.
Maintain an emergency stop state or emergency stop released state until all the lamps light up



(4) When the execution of the insulation degradation detection function is completed, the execution results are displayed as shown in the following screen



(5) Press the [END] soft key to close the window. Tue dedicated screen for the insulation degradation detection function will be updated.

NOTE

The execution of the insulation degradation detection function will be suspended in the following state, and the [RETRY] soft key.

- When an emergency stop state or emergency stop released state was not maintained until all the lamps light up.
- When the [CANCEL] soft key was pressed

Pressing the [RETRY] soft key shows the guidance again, enabling the insulation degradation detection function to be executed.

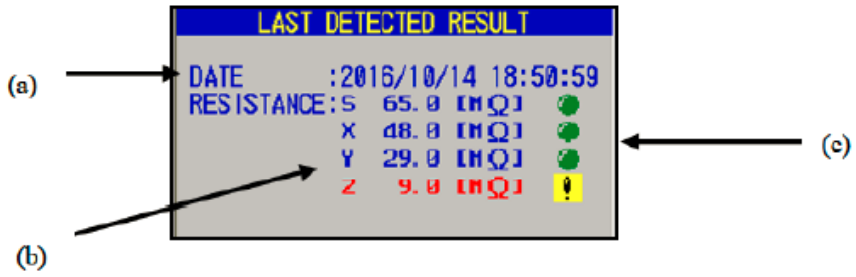
Press the [END] soft key to close the "EXECUTE INSULATION DEGRADATION DETECTION" window.

47.3.2 *CHECKING EXECUTION RESULTS OF INSULATION DEGRADATION DETECTION FUNCTION*

You can check the execution results of the insulation degradation detection function on the INSULATION DEGRADATION DETECTIONFUNCTION screen

Final detection results

Displays the date and time of the last execution of the insulation degradation detection function and the then measured insulation resistance value



(a) Date and time of detection

Displays the date and time of the last execution of the insulation degradation detection function.

(b) Insulation resistance value

Display the insulation resistance value for each axis measured when the insulation degradation detection function was executed the last time.

(c) *Axis insulation status*

Displays the insulation status of each axis with the following symbols

Normal (Tue insulation degradation detection function completed normally and no insulation degradation has completed)

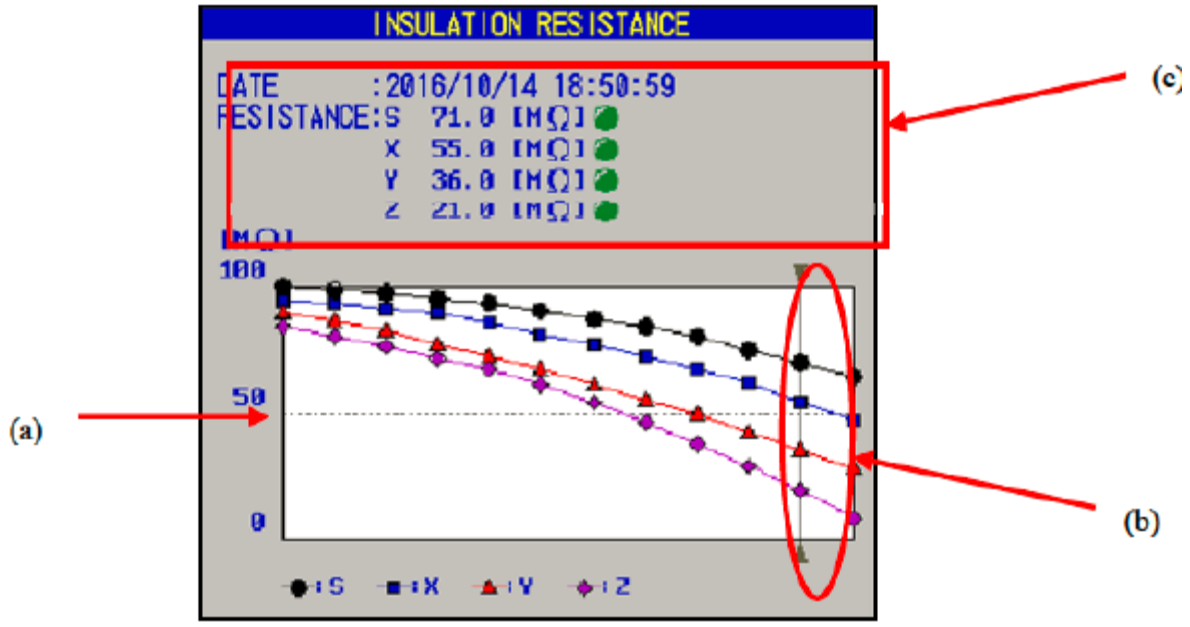
: Insulation degradation occurred (The insulation degradation detection function completed normally and insulation degradation has occurred)

: Measurement failed (The insulation degradation detection function did not complete normally)

: Measurement not performed (The insulation degradation detection function was not executed)

Changes of insulation resistance value

Saves the execution results of the insulation degradation detection function for up to 12 times and shows them in a graph



(a) Graph of insulation resistance value

Display the insulation resistance value for each axis measured by the insulation degradation detection function in a line graph.

(b) Cursor

Tue cursor appears on the graph and can be moved with the cursor keys <+-> + <- >.

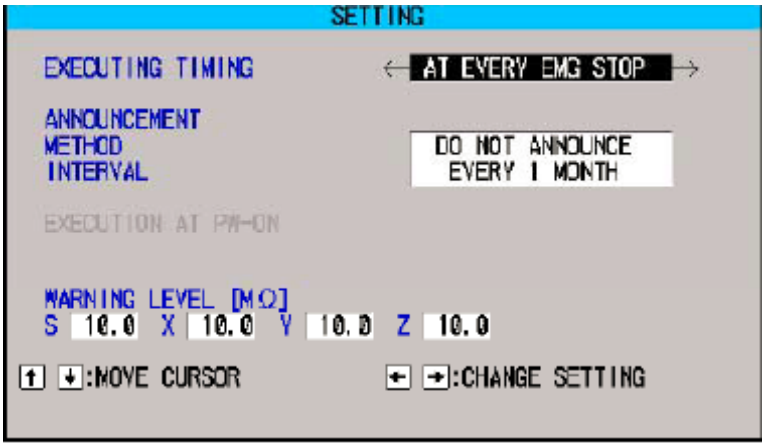
(b) Date and time of detection, insulation resistance value

Display the date and time when the insulation degradation detection function was executed and insulation resistance value for each axis for the point over which the cursor is placed on the graph. Tue content is the same as that for the "Last detection result" item.

47.2.1 Settings Related to the Execution of the Insulation Degradation Detection Function

You can set the conditions to execute the insulation degradation detection function and save detection results in the "Execute insulation degradation detection" item on the "SETTINGS" window

Select the "Execute insulation degradation detection" item using the cursor keys <j> + <J>, then change the setting using the cursor keys <- > + <- >.



Always execute at emergency stop (initial setting)

The insulation degradation detection function is executed and execution results are saved at the emergency stop state even on screens other than the INSULATION DEGRADATION DETECTION FUNCTIN screen.

For details on the execution of the insulation degradation detection function on non-dedicated screen, see "47.8.2 Execution on non-dedicated screens."

Execute on this screen only

The insulation degradation detection function can be executed and execution results can be saved only on the INSULATION DEGRADATION DETECTION FUNCTIN screen.

The function will not be executed on other screens, and the date and time of execution and insulation resistance value will not be saved

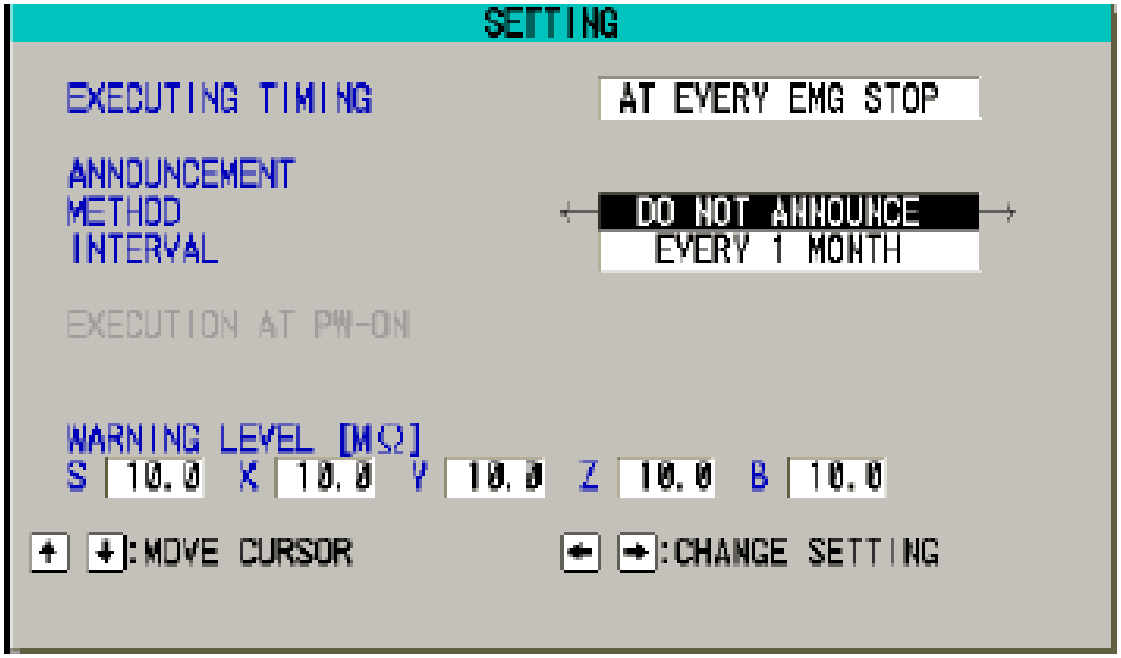
47.3 NOTIFYING EXECUTION OF INSULATION DEGRADATION DETECTION FUNCTION

By displaying an alarm or operator message at a certain interval the need to execute the insulation degradation detection function can be notified so that the function is executed at regular intervals.

47.3.1 Settings

The conditions to display an alarm or operator message can be set by changing the "Notification method" and "Notification interval " items under the "Notification of execution of detection item on the "SETTINGS" window.

Select the "Notification method" or "Notification interval "item under "Notification of execution of detection" using the cursor keys <j> + <J>, thenchai1ge the setting using the cursor keys <- > + <- >.



Notification method

Select the method to notify the execution of the insulation degradation detection function.

Not to notify (initial setting)

The execution of the insulation degradation detection function will not be notified with an alarm or operator message.

Notify with alarm

Notifies the time of executing the insulation degradation function with an alarm

Notify with message

Notifies the time of executing the insulation degradation function with an operator message.

Notification interval

Select the inte1val to display an alarm or operator message.
You can select the interval from "every one month" (initial setting), "**every three months**," "**every six months**," and "**every one year**"

> **NOTE**
> - Here, one month is considered to have 730 hours.

47.3.2 Specifications

When the time set in the "Notification inte1val" has passed after the last execution of the insulation degradation detection function, an alarm or operator message is displayed. When "Notify with alarm" is selected, "EX1098 Execute the insulation degradation detection function" is displayed. When "Notify with message" is selected, "MS2057 Execute the insulation degradation detection function" is displayed.

> **NOTE**
> - When "Notify with alarm" is selected, an alarm will not be displayed while the program is running. When the program ends, an alarm is displayed.
> - You can delete the alarm using the <RESET> key.

47.4 SWITCHING SCREEN WHEN POWER IS SWITCHEDON

This function cannot be used on the ROBODRILL *a-DiB* series.

47.5 WARNING LEVEL

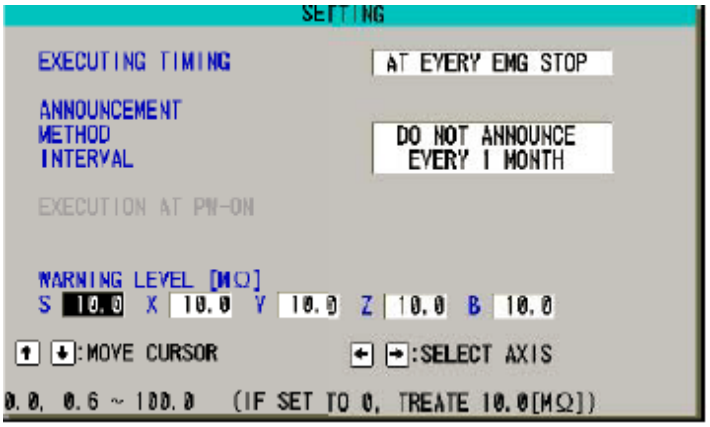
The insulation resistance value for dete1mining that insulation degradation has occurred is set.
If the insulation resistance value measured by the insulation degradation detection function is below the waiting level, insulation degradation is determined to have occurred.

47.5.1 Warning Level Setting

The warning level for each axis can be set in the "Warning level" item on the "SETTINGS" window.

(1) Select the "Warning level" item using the cursor keys <i> + <J>.
(2) Select the axis to change the setting using the cmsor keys<+-> + <- >.

(3) Enter the number using the numeric keypad on the operator's panel, then press the <INPUT> key to change the warning level.



47.2 OUTPUTTING EXECUTION RESULTS

You can output the saved execution results of the insulation degradation detection function to a memory card or USB memo1y.

1. Press the [OUTPUT] soft key. The following window appears
2. Enter the output file name, and press the [EXECUTE] soft key. A file containing the execution results is created in a mem01y card or USB memo1y under the file name you entered (the default name: "**LDMDAT.CSV**")

LEAKAGE DETECTION DATA								
No.	DATE	TIME	SPINDLE	SERVO X	SERVO Y	SERVO Z	ADD.4	ADD.5
1	2012/01/07	16:34:25	99.5	99.0	97.5	95.0	92.5	92.5
2	2012/02/02	3:55:51 PM	97.5	98.5	95.0	90.0	90.0	89.5
:	:	:	:	:	:	:	:	:
12	2013/01/31	4:24:19 PM	70.0	68.0	49.5	35.0	60.0	55.0
END								

(a) (b) (c) (d) (e)

(a) The order of executing the insulation degradation detection function is shown.
(b) The date, time, and hour when the insulation degradation detection function was executed are shown.
(c) The insulation resistance value of the main axis is shown.
(d) The insulation resistance value of the servo motor for each axis is shown.
The insulation resistance value of the additional axis is shown

47.6 SUPPLEMENT

47.6.1 Action to be Taken when Insulation Degradation is Detected

If insulation degradation is detected, measure the insulation resistance value of the motor and power cable separately using a meg ohmmeter to ve1ify whether insulation degradation occurred on the motor side or power cable side.
Then, replace the motor or power cable in which insulation degradation occurred

47.6.2 Execution on a Non-dedicated Insulation Degradation Detection Function Screen

If "Always execute at emergency stop" is set for the "Execute insulation degradation detection" item on the "SETTINGS" window on the INSULATION DEGRADATION DETECTION FUNCTIN screen, the insulation degradation detection function can be executed in the following steps on a screen other than the INSULATION DEGRADATION DETECTION FUNCTIN screen.

(I) In the emergency stop state, operate the EMERGENCY STOP button on the operator's panel to release the state, and keep the state for five seconds or more.

> **NOTE**
> - This operation charges the DC link of the servo amplifier.
> - This operation is not required if it is not in an emergency stop state.

(2) Press the EMERGENCY STOP button on the operator's panel to set an emergency stop state, and keep the state for about six seconds or more. The insulation resistance value of each axis is measured and insulation degradation is checked.

> **& WARNING**
> A high voltage is applied to the motor and cable for six seconds during the execution of the insulation degradation detection function. Do not touch the motor, cable between the motor and amplifier, and servo amplifier during the period to avoid an electrical shock.

> **NOTE**
> For high accuracy detection, set the state to the emergency stop state after the motor stoos.

47.6.3 Checking Execution Results on a Non-dedicated Insulation Degradation Detection Function Screen

On machines in which the dedicated INSURATION DEGRADATION DETECTION screen cannot be used, execution results can be checked on the following screens.

NOTE

- You cannot use these screen if "Execute on this screen only" is set for the "Execute insulation degradation detection" item on the "SETTINGS" window of the dedicated INSULATION DEGRADATION DETECTION FUNCTIN screen.
- On these screens, the execution results of the insulation degradation detection

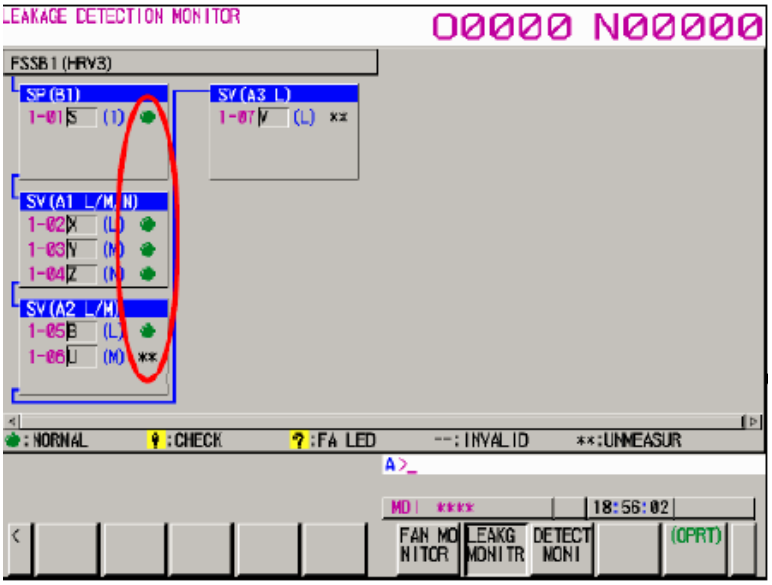
Checking if Insulation Degradation Has Occurred or Not

(I) When the CNC screen is being displayed, press the <SYSTEM> key.

(2) Press the [MAINTENANCE MONITOR] soft key. If this soft key is not displayed, press the [>] soft key at the light end to switch the soft key.

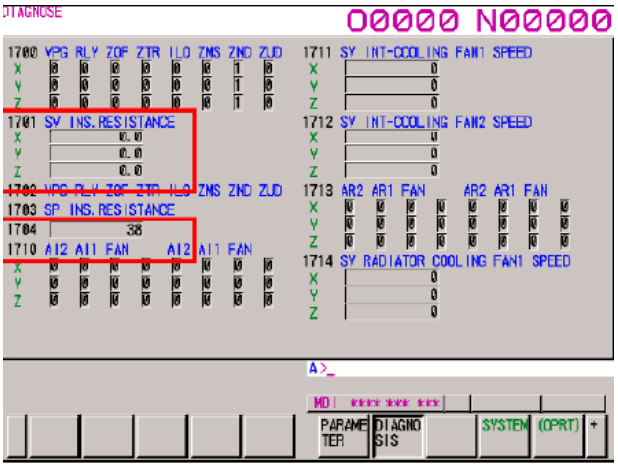
(3) Press the [INSURATION DEGRADATION MONITOR] soft key. The following INSURATION DEGRADATION MONITOR screen appears.

Symbols are used to indicate the insulation status of axes where small circles are shown



Checking the insulation resistance value

Check diagnosis info1mation DGN.1701 (servo axes) and DGN.1703 (main axis) (unit: MQ) for measured insulation resistance values



However,

- *This function will not generate the alarm*

- *The detection of the "Insulation Deterioration is a warning", and CNC does not have the function to generate the alarm.*
- *The ROBDRILL provides the Insulation Deterioration output signals.*
- *Please add the alarm message by using this signal on the customer side*
- *Below signal details available in Custom PMC Manual*

R 1432.2	SI82_2	Insulation deterioration detection	ON when an insulation deterioration of any axis is detected.